PER SCHOLAS

Report

SBA 930 - Evaluating LLMs for Financial Forecasting and Risk Analysis

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Overview

In this assessment, I evaluated how Large Language Models can be used for financial forecasting and risk analysis by comparing ChatGPT and Claude across several financial scenarios. The lab focused on market trends, economic indicators, commodity forecasting, and financial risk identification. I examined the accuracy, relevance, and reliability of each model's responses while also considering their strengths, limitations, and potential sources of error. The assessment also explored ways to improve LLM performance through better prompting, reliable data sources, human oversight, and responsible use in financial decision-making.

For the resubmission, I added a reproducible retrospective backtest to quantitatively evaluate the two models. Both models produced baseline and optimized forecasts for the S&P 500, December WTI crude oil, and December U.S. CPI using only historical values supplied through 2023. The forecasts were then compared with held-out 2024 actual outcomes using absolute error and absolute percentage error (APE). Mean absolute percentage error (MAPE) was used to compare aggregate model performance and measure the effect of prompt optimization.

GitHub Repository
The datasets, project setup, exact prompts, complete model responses, evaluation code, calculations, and generated result files are available at the public repository below.

github.com/ZachariasBraetyllo-Sec/SBA_930_Zacharias_Braetyllo

This expanded report preserves the original qualitative comparison while adding measurable forecast validation and an implemented optimization experiment. The quantitative results are intended as a historical evaluation exercise rather than evidence that either LLM can reliably predict future financial markets.

Steps

1. LLM Evaluation for Financial Forecasting

Prompt 1: Using historical S&P 500 performance as the financial dataset, analyze how recent economic policies could affect the stock market over the next 6 to 12 months. Identify the main economic factors that could influence market performance, explain whether each factor is likely to have a positive or negative effect, and describe the uncertainty involved in the forecast. Do not present the forecast as guaranteed.

Evidence: See Appendix A, Figures A1-A2.

Comparison: Both ChatGPT and Claude identified similar major factors affecting the S&P 500, including interest rates, inflation, energy prices, corporate earnings, AI investment, and trade policy. Both models also emphasized that the forecast was uncertain and should not be treated as guaranteed.

ChatGPT gave a more concise and structured response. It clearly separated the main economic factors and explained whether each was likely to have a positive, negative, or mixed effect on the market. Its main strength was clarity and ease of interpretation. However, it provided less historical context and fewer quantitative details.

Claude provided a much more detailed analysis and included historical return patterns, Treasury yields, inflation figures, oil prices, tariff policy, election uncertainty, and several possible market scenarios. Its response showed stronger depth and risk analysis, but it was also much longer and included several approximate historical figures that Claude itself noted should be verified.

Overall, Claude performed better in depth and risk discussion, while ChatGPT performed better in clarity and conciseness. Both responses were relevant to the prompt, but neither should be treated as a reliable prediction because financial markets are influenced by changing economic and political conditions.

Observation: This comparison showed that two LLMs can reach similar conclusions while presenting them very differently. Claude provided more detailed historical and economic context, while ChatGPT produced a more focused summary. This demonstrates why LLM output should be evaluated not only for the amount of information provided, but also for accuracy, relevance, and how clearly uncertainty is communicated.

Prompt 2: Using historical crude oil price data as the financial dataset, analyze recent price trends and predict the most likely direction of crude oil prices over the next 6 to 12 months. Identify the main factors that could push prices higher or lower, explain how historical patterns inform the forecast, and describe the uncertainty involved. Do not present the forecast as guaranteed.

Evidence: See Appendix A, Figures A3-A4.

Comparison: Both ChatGPT and Claude concluded that crude oil prices are more likely to decline from current elevated levels over the next 6 to 12 months, while remaining highly volatile. Both identified geopolitical conflict, supply disruptions, inventories, OPEC+ policy, production recovery, and global demand as major factors affecting future prices.

ChatGPT gave a concise explanation of the main forces pushing prices higher or lower. It emphasized supply disruptions, low inventories, future production recovery, OPEC+ policy, and economic demand. Its main strength was clarity, because the response made the directional effect of each factor easy to understand. However, it included less historical detail and fewer specific price scenarios.

Claude provided a much deeper analysis. It used historical oil shocks, approximate volatility ranges, agency forecasts, and three possible price scenarios to explain why prices could move significantly in either direction. It also discussed the limitations of historical comparisons and acknowledged that several figures were approximate or based on changing forecasts. Its main weakness was that the amount of detail made the response less concise and introduced more claims that would need verification.

Overall, Claude performed better in historical context, scenario analysis, and discussion of uncertainty, while ChatGPT performed better in clarity and concise explanation. Both produced similar directional forecasts, which increases confidence that the major risk factors were identified consistently, but neither response should be treated as a precise prediction.

Observation: This comparison showed that both models can identify similar trends from the same financial scenario while using different levels of detail. Claude relied more heavily on historical examples and numerical ranges, while ChatGPT focused on summarizing the most important market drivers. It also demonstrated that additional detail does not automatically make a forecast more reliable, especially when financial conditions can change rapidly and some figures require independent verification.

Historical Forecast Validation - S&P 500 and WTI Crude Oil

To add measurable forecast-accuracy evaluation, I conducted a retrospective backtest using values through 2023 as the model inputs and withheld the 2024 outcomes until scoring. Each model produced one baseline forecast and one optimized forecast for each indicator. The resulting point forecasts were compared with the held-out actual values.

S&P 500 Year-End 2024

Held-out actual result: 5,881.63

Model	Prompt	Forecast	Absolute Error	APE
ChatGPT	Baseline	5,250	631.63	10.74%
ChatGPT	Optimized	5,581	300.63	5.11%
Claude	Baseline	5,257	624.63	10.62%
Claude	Optimized	5,008	873.63	14.85%

Result: ChatGPT's optimized prompt substantially reduced S&P 500 forecast error, while Claude's optimized prompt increased error. This shows that more structured prompting can improve a forecast in some cases but does not guarantee better numerical accuracy.

December WTI Crude Oil Monthly Average

Held-out actual result: $70.12 per barrel

Model	Prompt	Forecast	Absolute Error	APE
ChatGPT	Baseline	$72.50	$2.38	3.39%
ChatGPT	Optimized	$72.00	$1.88	2.68%
Claude	Baseline	$73.00	$2.88	4.11%
Claude	Optimized	$71.00	$0.88	1.25%

Result: Both models improved under the optimized prompt for WTI. Claude produced the most accurate WTI forecast in the experiment, with a 1.25% absolute percentage error.

2. Economic Trend Forecasting

Prompt: Using historical U.S. inflation data, analyze recent inflation trends and forecast the likely direction of inflation over the next 6 to 12 months. Identify the main economic factors that could cause inflation to rise or fall, explain how historical patterns inform the forecast, and describe the uncertainty involved. Do not present the forecast as guaranteed.

Evidence: See Appendix A, Figures A5-A6.

Comparison: Both ChatGPT and Claude forecast that U.S. inflation is likely to ease over the next 6 to 12 months, although neither expects a smooth decline. Both identified energy prices, Federal Reserve policy, labor-market conditions, tariffs, and supply pressures as major influences on inflation.

ChatGPT gave a more concise explanation of the main inflation drivers and clearly labeled whether each factor was likely to push inflation higher or lower. Its main strength was readability and directness. However, it provided fewer numerical projections and less historical detail about previous inflation cycles.

Claude provided a much more detailed forecast. It separated headline inflation from core inflation, included several specific ranges, used historical energy-shock examples, and described multiple possible scenarios. It also discussed base effects and differences between CPI and PCE measures. Its main weakness was that the amount of detail introduced more approximate figures and assumptions that would need independent verification.

Overall, Claude performed better in depth, historical context, and scenario analysis, while ChatGPT performed better in clarity and concise interpretation. Both models produced similar overall conclusions, which suggests that the main inflation risks were identified consistently, but the exact numerical forecasts remain uncertain.

Observation: This comparison showed that LLMs can be useful for organizing economic indicators into a forecast, but the quality of the result depends heavily on how much detail and uncertainty the model communicates. Claude provided a more technical analysis, while ChatGPT made the major trends easier to understand quickly. The exercise also showed why economic forecasts should be compared with actual historical data and official sources rather than accepted at face value.

Historical Forecast Validation - U.S. Inflation

For the quantitative backtest, both models were given December year-over-year CPI values from 2020 through 2023 and asked to forecast December 2024 without using 2024 information. The held-out actual December 2024 CPI year-over-year rate was 2.9%.

Model	Prompt	Forecast	Absolute Error	APE
ChatGPT	Baseline	2.5%	0.4 pp	13.79%
ChatGPT	Optimized	2.8%	0.1 pp	3.45%
Claude	Baseline	2.4%	0.5 pp	17.24%
Claude	Optimized	2.7%	0.2 pp	6.90%

Both models became more accurate after prompt optimization. ChatGPT's CPI error fell from 13.79% to 3.45%, while Claude's fell from 17.24% to 6.90%. The optimized ChatGPT forecast of 2.8% was closest to the held-out actual result.

Accuracy Metric

Absolute percentage error (APE) was calculated as the absolute difference between forecast and actual divided by the actual value, multiplied by 100. This makes the forecast errors comparable within each indicator. Mean absolute percentage error (MAPE) was then calculated across the three forecast tasks for each model and prompt condition.

Why this matters

The original report compared the quality and reasoning of the two models. The backtest adds a measurable outcome: how far each point forecast was from an actual historical result. This allows the qualitative observations to be tested against numerical performance rather than relying only on response quality.

3. Risk Identification and Mitigation

Prompt: Using the financial conditions discussed in the previous analyses, identify the most important risks related to market volatility, credit risk, and liquidity risk over the next 6 to 12 months. Explain how each risk could affect financial institutions or investors, rank the risks by severity, and propose practical mitigation strategies for each. Include uncertainty and avoid presenting any risk estimate as guaranteed.

Evidence: See Appendix A, Figures A7-A8.

Comparison: Both ChatGPT and Claude identified market volatility, credit risk, and liquidity risk as important concerns over the next 6 to 12 months. Both also emphasized that the risks are interconnected and could worsen together if interest rates remain high, energy prices stay elevated, or economic conditions weaken.

ChatGPT ranked market volatility as the highest near-term risk, followed by credit risk and liquidity risk. Its response was concise and easy to follow, with practical mitigation strategies such as diversification, stress testing, stronger credit screening, cash reserves, and liquidity planning. Its main strength was clarity, but it stayed at a broader level and did not explore specific financial-market stress scenarios in much depth.

Claude provided a more detailed and granular risk assessment. It ranked Treasury-market liquidity stress first, followed by private-credit stress, equity drawdowns, oil shocks, credit deterioration, and bank-funding strain. It also explained who could be affected by each risk and proposed separate mitigation strategies for institutions and investors. Its main strength was depth, although the response relied on more specific market figures and assumptions that would require verification.

Overall, Claude performed better in risk segmentation, scenario detail, and institutional analysis, while ChatGPT performed better in clarity, simplicity, and direct prioritization. Both models identified useful mitigation strategies, but Claude's additional detail made its response more comprehensive while also increasing the amount of information that would need to be validated.

Observation: This comparison showed that LLMs can identify similar categories of financial risk while differing significantly in how deeply they analyze them. ChatGPT focused on broad risk categories and straightforward mitigation strategies, while Claude broke the risks into more specific market and institutional scenarios. The quantitative backtest reinforces the need for the same safeguards: model error varied by task and prompt, so stress testing, source verification, and human review remain necessary before LLM output is used in financial decisions.

4. Optimization and Enhancement

Prompt: Using the weaknesses identified in the previous financial forecasting and risk-analysis tasks, recommend methods to improve the accuracy, reliability, and usefulness of LLMs in financial applications. Explain how techniques such as better prompt design, verified financial data, retrieval-augmented generation, human review, and uncertainty reporting could improve performance. Identify the limitations that would still remain even after these improvements.

Evidence: See Appendix A, Figures A9-A10.

Comparison: Both ChatGPT and Claude recommended improving financial LLM performance through better prompt design, verified data sources, retrieval-augmented generation, human review, and explicit uncertainty reporting. Both also agreed that these methods can improve reliability but cannot eliminate the unpredictability of financial markets.

ChatGPT focused on the main improvement areas in a concise and practical way. It explained how clearer prompts, trusted financial data, RAG, human oversight, and uncertainty reporting could reduce errors and make outputs easier to evaluate. Its main strength was clarity, although it stayed at a higher level and did not go deeply into implementation details.

Claude provided a more detailed optimization plan. It identified specific weaknesses from the earlier tasks, such as relying on approximate historical figures, conflicting sources, outdated data, and untested scenario ranges. It then connected those weaknesses to specific solutions, including authoritative data sources, source ranking, quantitative models, backtesting, calibration tracking, structured evaluation, and monitoring.

Overall, Claude performed better in diagnosing specific weaknesses and proposing technical improvements, while ChatGPT performed better in clear and concise explanation of the major optimization strategies. Both models agreed that LLMs should support financial analysis rather than make final decisions independently.

Observation: This comparison showed that improving an LLM involves more than changing the wording of prompts. Reliable financial use also requires better data pipelines, source verification, quantitative tools, evaluation methods, and human oversight. Even after these improvements, uncertainty and unexpected market events would remain important limitations. This reinforces the idea that LLMs are most effective as analytical support tools rather than standalone forecasting systems.

Implemented Optimization Experiment

To move the optimization section from recommendations to implementation, I ran a controlled before-versus-after prompt experiment. The baseline prompts asked each model to make a point forecast from the supplied historical values. The optimized prompts required each model to identify direction and volatility, separate observations from assumptions, state a major uncertainty, and produce a conservative final forecast.

What changed in the optimized prompts

- Verified historical inputs were supplied directly in the prompt instead of asking the model to retrieve values independently.
- The model was explicitly instructed not to use or mention information from the held-out year.
- The prompt required observations and assumptions to be separated so unsupported reasoning was easier to identify.
- The model was required to state uncertainty and return one final numeric point forecast for scoring.

Before-versus-after results

Model	Baseline MAPE	Optimized MAPE	Improvement	Relative Improvement
ChatGPT	9.31%	3.75%	5.56 points	59.72%
Claude	10.66%	7.67%	2.99 points	28.05%

Optimization improved aggregate accuracy for both models. ChatGPT showed the larger improvement, reducing MAPE by 59.72%. Claude improved by 28.05%. However, Claude's S&P 500 forecast became less accurate after optimization, increasing from 10.62% error to 14.85%. This negative task-level result is important because it demonstrates that prompt optimization should be measured rather than assumed to work.

The experiment therefore supports a narrower conclusion: structured prompting improved overall performance in this small backtest, but the effect depended on the model and forecast task. Quantitative evaluation remains necessary after any optimization change.

Quantitative Model Comparison

The table below summarizes all twelve forecasts used in the evaluation. Lower APE indicates a forecast closer to the held-out actual result.

Model	Prompt	Metric	Forecast	Actual	APE
ChatGPT	Baseline	S&P 500	5,250	5,881.63	10.74%
ChatGPT	Optimized	S&P 500	5,581	5,881.63	5.11%
Claude	Baseline	S&P 500	5,257	5,881.63	10.62%
Claude	Optimized	S&P 500	5,008	5,881.63	14.85%
ChatGPT	Baseline	WTI	$72.50	$70.12	3.39%
ChatGPT	Optimized	WTI	$72.00	$70.12	2.68%
Claude	Baseline	WTI	$73.00	$70.12	4.11%
Claude	Optimized	WTI	$71.00	$70.12	1.25%
ChatGPT	Baseline	CPI	2.5%	2.9%	13.79%
ChatGPT	Optimized	CPI	2.8%	2.9%	3.45%
Claude	Baseline	CPI	2.4%	2.9%	17.24%
Claude	Optimized	CPI	2.7%	2.9%	6.90%

Key quantitative findings

- ChatGPT had the lower aggregate MAPE in both conditions: 9.31% baseline and 3.75% optimized.
- Claude achieved the best individual WTI result after optimization, with 1.25% APE.
- ChatGPT achieved the best optimized S&P 500 and CPI results, with 5.11% and 3.45% APE respectively.
- The optimized condition improved five of the six model-by-indicator comparisons; the exception was Claude’s S&P 500 forecast.

Technical Reproducibility

The project was implemented as a reproducible Python evaluation rather than a screenshot-only comparison. The repository contains the historical input data, held-out outcomes, extracted model forecasts, exact prompts and responses, executable scoring code, generated calculations, and environment metadata.

Repository: https://github.com/ZachariasBraetyllo-Sec/SBA_930_Zacharias_Braetyllo

Artifact	Purpose
data/historical_inputs.csv	Historical values supplied to the models.
data/actual_results.csv	Held-out 2024 actual outcomes used only for scoring.
data/model_forecasts.csv	All 12 extracted baseline and optimized forecasts.
artifacts/prompts.md	Exact baseline and optimized prompt templates.
artifacts/model_responses.md	Complete ChatGPT and Claude forecast responses.
src/evaluate.py	Reproducible APE, MAPE, and optimization calculations.
results/forecast_evaluation.csv	Forecast-level actuals, errors, and APE.
results/model_summary.csv	Model-level MAPE by prompt condition.
results/optimization_comparison.csv	Before-versus-after optimization comparison.
pyproject.toml / uv.lock	Project and Python environment metadata.

Data sources and evaluation design

The historical evaluation inputs were based on S&P 500 year-end values, WTI December monthly-average prices, and U.S. CPI December year-over-year inflation. The repository documents the corresponding official-data sources, including FRED/S&P Dow Jones, FRED/EIA, and the U.S. Bureau of Labor Statistics. The held-out 2024 actual values used for scoring were 5,881.63 for the S&P 500, $70.12 for December WTI, and 2.9% for December CPI year-over-year.

Limitation of the retrospective backtest

The prompts explicitly instructed the models to use only the supplied pre-2024 values. However, because the models were queried after 2024, it is not possible to prove that their training or internal knowledge contained no exposure to 2024 historical outcomes. The backtest therefore measures behavior under constrained prompts, not a perfectly isolated real-time 2023 forecasting environment. This limitation should be considered when interpreting the numerical accuracy.

Conclusion

This assessment showed that LLMs can be useful tools for financial forecasting and risk analysis, but their value depends heavily on how they are used. Both ChatGPT and Claude were able to identify similar economic trends, market risks, and mitigation strategies, although they differed in presentation and depth. ChatGPT generally produced clearer and more concise responses, while Claude provided more detailed historical context, scenario analysis, and technical discussion.

The expanded quantitative backtest added measurable evidence to that qualitative comparison. Across the three held-out 2024 forecasting tasks, ChatGPT produced a baseline MAPE of 9.31% and Claude produced 10.66%. After the structured prompt optimization was implemented, ChatGPT improved to 3.75% MAPE and Claude improved to 7.67%. This represented relative improvements of 59.72% and 28.05%, respectively. The results also showed that optimization was not universally beneficial, because Claude's optimized S&P 500 forecast became less accurate even while its overall MAPE improved.

The comparison also highlighted important limitations. LLMs can rely on outdated or conflicting information, make unsupported assumptions, and present uncertain analysis in a confident way. Their performance can be improved through better prompts, verified financial data, retrieval-augmented generation, quantitative tools, uncertainty reporting, and human review. However, those improvements should be tested with measurable evaluation rather than assumed to improve every task.

Overall, LLMs are most effective when used as analytical support systems rather than independent financial decision-makers. They can help organize information, identify patterns, and generate useful scenarios, but their outputs should still be verified against reliable sources and reviewed by qualified humans before being used in high-impact financial decisions. The GitHub repository provides the datasets, exact model outputs, code, calculations, and evaluation artifacts needed to reproduce the quantitative portion of this assessment.

Appendix A: Prompt and Response Screenshots

Each page below pairs the ChatGPT and Claude responses used for the corresponding qualitative comparison.

Figure A1. ChatGPT response - Section 1, Prompt 1: S&P 500

A1 ChatGPT SP500

Using historical S&P 500 performance as the financial dataset, the next 6 to 12 months appear to have both positive growth potential and meaningful downside risk. The S&P 500 recently reached record highs, supported by strong corporate earnings and continued investment in technology and AI, but higher interest rates, inflation pressures, and elevated bond yields could limit further gains.

Interest rates: Likely negative to mixed. The Federal Reserve raised rates in September 2026, and additional increases remain possible if inflation stays elevated. Higher rates can raise borrowing costs for companies and consumers while making bonds more attractive relative to stocks. However, weaker economic data could lead the Fed to pause further increases, which would be more supportive of equities.

Inflation and energy prices: Likely negative if pressures persist. Oil prices above $100 and other inflationary shocks could increase costs for businesses and consumers. Persistent inflation could also encourage the Federal Reserve to maintain tighter monetary policy for longer.

Corporate earnings and AI investment: Likely positive. Strong earnings growth, particularly in technology and energy, has helped support the S&P 500. Continued investment in AI infrastructure could provide additional growth, although heavy dependence on a small group of high-performing sectors creates concentration risk.

Fiscal and trade policy: Mixed. Policies that encourage business investment or economic growth could support corporate profits, while tariffs or other trade restrictions could increase input costs and contribute to inflation. These effects may vary significantly by industry.

Overall, I would describe the 6-to-12-month outlook as cautiously positive but highly uncertain. Strong earnings and technology investment could support further gains, while high interest rates, inflation, energy costs, and geopolitical or policy changes create meaningful downside risks. Historical S&P 500 performance can provide useful context, but it cannot reliably predict how the market will respond to new economic conditions, so this forecast should be treated as a scenario analysis rather than a guaranteed prediction.

Figure A2. Claude response - Section 1, Prompt 1: S&P 500

Short version: the main drivers I see are earnings strength and AI spending (positive), against tighter Fed policy, high oil prices, and rising bond yields (negative). History says a positive 12-month return is more likely than not, but the range of outcomes is wide, and the current setup has more tail risk than average. This is an analysis of risks, not a prediction or financial advice.

Where things stand (as of Oct 6, 2026)

The S&P 500 closed at a record 7,818.93. That is happening alongside a hawkish policy backdrop:

- The Fed raised rates 25 bps in a 12-0 vote, its first hike since 2023, to 3.75%-4.00%. Sixteen of eighteen officials expect one more hike this year.
- The 10-year Treasury yield has been above 5%. At one point it reached levels last seen in 2002, and the 30-year yield topped 5.7%.
- August CPI was 3.4% year over year, with core at 2.4%, and energy up 16.3%. Most of the inflation is energy-driven.
- Brent crude is around $101 a barrel, roughly 54% above a year ago.

Historical baseline

These figures are from my general knowledge of long-run S&P 500 data. They are approximate, so verify them before relying on them:

- The S&P 500 has been up in roughly 70-75% of calendar years and rolling 12-month windows, with long-run total returns near 10% a year.
- Annual returns have a standard deviation of roughly 17-19%. A naive one-standard-deviation band around the average is about −8% to +28%, which is wide even before today's specific risks.
- 10% pullbacks happen roughly once a year on average, and 20%+ bear markets every few years.
- Midterm years have historically been the weakest part of the presidential cycle, with larger intra-year drawdowns. The 12 months after midterms have historically been strong. That pattern is based on few observations.
- Early in past Fed hiking cycles, results were mixed (roughly flat to modestly positive in several, sharply negative in 2022).

Key factors

1. Fed policy and rates: negative. Higher rates reduce what investors will pay for future earnings, and 5%+ Treasuries compete with stocks, especially at elevated valuations. The Fed has signaled no change in 2027 and cuts only in 2028-2029. A weak jobs report recently eased fears of further hikes, so for now bad economic news can help stocks.

2. Oil and geopolitics (Iran conflict): negative, but the swing factor. Energy feeds inflation, which keeps the Fed hawkish. Yields have been tightly correlated with oil prices. If oil falls, the effect flips positive, as on Oct 6 when falling oil eased inflation worries and lifted stocks. The reverse is also possible.

3. Tariffs and trade policy: mixed and highly uncertain. After the Supreme Court struck down the IEEPA tariffs in February, the administration moved to Section 122 and then a new Section 301 tariff covering 60 economies, which faces legal challenges. Possible refunds of $166B+ in duties could help corporate cash flow. Tariffs and retaliation (e.g., Canada's new tariffs) still raise costs. Policy has changed direction several times, which makes this factor hard to model.

4. Fiscal policy and deficits: mixed. Talk of stimulus checks and a midterm-year fiscal push supports demand and sentiment. But investors are wary of government debt amid rising deficits, which keeps long-term yields high, a headwind for valuations.

5. AI investment and earnings growth: positive, with concentration risk. Goldman Sachs says 2026 gains have been driven by profit growth rather than higher valuations, with AI-infrastructure beneficiaries accounting for about half of earnings growth. The catch is that breadth has narrowed, with the AI trade driving the index. A stumble in a few large companies would hit the whole index.

6. Consumer and labor strength: negative to mixed. Consumer confidence recently fell to a 12-year low, and the jobs data is softening. That threatens growth but may also reduce pressure for more Fed hikes.

7. November midterms: uncertain. Historically, elections bring pre-vote volatility and post-vote relief, but the direction depends on the result and the policy changes it brings.

Uncertainty in the forecast

- **Wide outcome range.** A reasonable framing is three paths: (a) **upside:** oil eases, the Fed stops at one more hike, and AI earnings keep beating; (b) **base:** choppy, with modest gains and one or more 8-12% pullbacks; (c) **downside:** oil re-spikes, yields climb further, and a narrow, richly valued market de-rates. I wouldn't assign precise probabilities to these.
- **History is an imperfect guide.** A hiking Fed, 5% yields, an oil shock, and rebuilt tariff laws together have few close precedents.
- **Policy is unpredictable.** Tariff authorities, Fed leadership, and fiscal plans have all shifted within the year, and any could shift again.
- **Elevated starting point.** Starting from record highs doesn't predict a decline. But with valuations high and breadth narrow, there is less cushion if something goes wrong.
- **Data quality.** I relied on news and market summaries, and analysts' 2026 targets (typically 7,300-8,000) were set before the oil shock and Fed hikes, so treat them cautiously.

For decisions, a financial advisor can weigh this against your time horizon and risk tolerance. I'm not a financial advisor.

Figure A3. ChatGPT response - Section 1, Prompt 2: Crude Oil

A3 ChatGPT Crude Oil

Using historical crude oil price data, the most likely direction over the next 6 to 12 months is downward from current elevated levels, but with continued volatility. Brent crude is currently above $100 per barrel and WTI is near $90, largely because of supply disruptions, geopolitical conflict, and very low global inventories.

Supply disruptions and geopolitical conflict: Likely positive for prices. Ongoing instability in the Middle East and Ukraine has restricted oil flows and increased transportation risks. Any further disruption to major producers or shipping routes could push prices higher quickly.

Global inventories: Likely positive in the short term. Oil inventories have fallen sharply, leaving the market with less protection against additional supply shocks. Low inventories can create a higher price floor and increase volatility.

Production recovery and higher supply: Likely negative over the longer term. The EIA expects disrupted Middle Eastern production and exports to gradually recover through 2027. It forecasts Brent falling to about $87 per barrel by the second quarter of 2027 as supply improves, with further declines possible as inventories rebuild.

OPEC+ production policy: Mixed. OPEC+ has kept its near-term production targets unchanged, which supports prices while the market remains tight. Future production increases could place downward pressure on prices if supply grows faster than demand.

Economic growth and demand: Mixed to negative. Strong global growth could maintain demand for oil, while slower economic activity would reduce consumption and put downward pressure on prices.

Historical patterns show that crude oil can move sharply when supply is disrupted, but high prices often encourage additional production, alternative supply routes, and lower demand. This can eventually pull prices back down.

Overall, I would expect crude oil prices to remain elevated and volatile in the near term before gradually declining over the next 6 to 12 months if production and trade flows recover. However, geopolitical events could quickly change that outlook, so the forecast should be treated as a scenario rather than a guaranteed prediction.

Figure A4. Claude response - Section 1, Prompt 2: Crude Oil

Brent crude is more likely to end the next 6 to 12 months below today's roughly $101 than above it, but the odds favor "somewhat lower and very volatile" rather than a clean decline. This is an analysis of risks, not a prediction or financial advice.

Recent trend

- Brent was around $101 on Oct 6, about 54% above a year ago, when it traded near $66.
- Brent spiked to about $118 in late March after the Strait of Hormuz was effectively closed, fell to around $70 by July 1, rebounded above $100 in late July, and traded between $87 and $97 in August. The IEA called it the largest supply disruption in oil market history.
- Hormuz-related shut-in production averaged 4.8 million barrels a day in September, down from 5.8 million, but the EIA now expects Brent to average $105 in Q4 2026.
- WTI is lower, near $90, so the Brent-WTI gap is unusually wide.

How history informs the forecast

These are from my general knowledge of past oil cycles, so treat the figures as approximate:

- **Supply-shock spikes tend to fade.** In 1990 crude roughly doubled after Iraq invaded Kuwait, then fell back within months. In 2022 Brent peaked near $128 in March and was far lower a year later. Prices above long-run costs trigger demand destruction, more non-OPEC output, and reserve releases. The EIA expects U.S. crude output to rise from 13.8 to 14.3 million barrels a day in 2027.
- **Oil is very volatile.** Annual swings have a standard deviation of roughly 30-35%. Around $101 that implies a one-standard-deviation band of about $65-$135, which is a statistical range, not a forecast.
- **Forecasters often miss.** The EIA's October 2025 outlook predicted a 2026 Brent average of $52. The September 2026 outlook puts it near $91. A large part of any forecast is guesswork about geopolitics.
- **Shocks sometimes persist.** In 1973-74 and 1979-80, prices stayed elevated for years after the initial spike. That is the scenario to take seriously if Hormuz stays contested.

Base case

Prices stay elevated near term (roughly $90-105 in Q4), then drift lower as shipping and production recover, ending the 12-month window somewhere in the $75-95 range. The EIA's September outlook sees Brent averaging $74 in 2027 as production rises and inventories rebuild. Reuters' September poll put the 2026 average at about $89, with individual forecasts ranging from $77 to $98. Analysts have been revising forecasts up, which is a reminder that the risks have lately skewed higher.

What could push prices higher

- **Hormuz disruption continues or worsens.** Iran has reportedly stepped up attacks on tankers, and Houthi missile activity continues.
- **Thin buffers.** The U.S. strategic reserve is near historic lows, so there is less cushion for another supply loss.
- **Tight inventories.** Saudi Aramco's CEO described stockpiles as scarily thin.
- **Slow OPEC+ response.** OPEC+ is unlikely to add much supply soon, because its output depends on normal flows through the strait.
- **Product market tightness.** Chinese refiners reportedly suspended October fuel exports.

What could push prices lower

- **Supply returns faster than expected.** The EIA expects most shut-in production back online by early 2027. A durable Iran deal, like the June MOU that briefly sent prices sharply lower, would likely have a big effect.
- **Weaker demand.** The Fed is hiking, consumer confidence recently hit a 12-year low, and high prices themselves curb consumption. A recession would be a strong bearish force.
- **Emergency supply.** G7 countries agreed to release 100 million barrels from reserves, and Saudi Arabia cut its Asian selling price as flows improved.
- **Non-OPEC growth** from the U.S., Brazil, Guyana, and others.

Scenarios (illustrative, not probabilities)

Scenario	Conditions	Brent in 6-12 months
Normalization	Hormuz flows recover, shut-ins restored by Q1 2027	~$65-80
Muddle-through (base)	Gradual recovery with periodic flare-ups	~$75-95
Renewed disruption	Re-closure or major infrastructure attacks, thin inventories	$110-135+

Uncertainty

- **Headline risk dominates.** Prices have moved $15-30 in weeks on single events, and a deal or attack can arrive without warning.
- **Weak precedents.** A shock this large, combined with depleted reserves and a hiking Fed, has few close historical analogs.
- **Data limits.** I used news and agency summaries, some of which conflict on exact figures. Check current quotes and the latest EIA outlook before relying on any number here.
- **Skewed risks.** The most likely path is lower, but a spike has a larger effect on the economy and markets than an equally likely decline.

I'm not a financial advisor, and decisions about trading or hedging should consider your own circumstances.

Figure A5. ChatGPT response - Section 2: U.S. Inflation

A5 ChatGPT Inflation

Using historical U.S. inflation data, the most likely direction over the next 6 to 12 months is gradually lower inflation, but with meaningful risk of temporary increases. The latest available CPI data show headline inflation at 3.4% year over year in August 2026, while core inflation was lower at 2.4%. Gasoline prices were a major contributor to the monthly increase.

Energy prices: Likely positive for inflation in the short term. Higher oil and gasoline prices can raise transportation and production costs, which can spread into other goods and services. Recent energy shocks have been one of the main reasons inflation remains above the Federal Reserve's 2% target.

Interest rates: Likely negative for inflation over time. The Federal Reserve raised rates in September 2026, which should help slow demand and reduce price pressures. However, policymakers remain divided about whether further increases are necessary, so the effect may be gradual rather than immediate.

Labor market conditions: Likely negative to mixed. September job growth slowed and wage growth remained moderate, reducing the risk of a wage-driven inflation cycle. A weaker labor market can lower consumer demand, but if employment remains strong enough to support spending, inflation may fall only slowly.

Supply chains and tariffs: Likely positive for inflation if pressures continue. Supply-chain stress increased in September, and tariffs can raise the cost of imported goods and materials. These factors could delay progress toward lower inflation even if demand weakens.

Historical patterns show that inflation often declines gradually after monetary policy becomes tighter, but supply shocks can interrupt that trend. The recent combination of elevated energy prices and tighter policy makes the outlook less predictable than a normal disinflation period.

Overall, I would expect U.S. inflation to trend modestly downward over the next 6 to 12 months, though probably not in a straight line. Energy prices, tariffs, supply disruptions, and future Federal Reserve decisions could keep inflation elevated or cause temporary increases. This forecast should be treated as a scenario analysis rather than a guaranteed prediction.

Figure A6. Claude response - Section 2: U.S. Inflation

Headline inflation looks likely to peak in the next few months and then ease through 2027, but the path depends heavily on oil. Core inflation looks stickier and will probably stay above the Fed's 2% target throughout. This is an analysis of risks, not a prediction or financial advice.

Recent trend

- August CPI was 3.4% year over year, with core at 2.4% and energy prices up 16.3%. The gap between headline and core shows this is mostly an energy shock, not broad-based inflation.
- The Fed's preferred gauge reads hotter. Core PCE was 3.0% and headline PCE 3.4%, with consumer inflation expectations at 3.6% and producer prices excluding food, energy and trade up 4.7%.
- The Cleveland Fed's nowcast points to September CPI around 3.6% year over year, and about 4.1% annualized for Q4, so headline inflation is probably still rising near term.
- The Fed hiked to 3.75%-4.00% in September. Its median projections now put core PCE at 3.4% for 2026, 2.5% for 2027, and about 2.1% for 2028, with fifteen of eighteen officials seeing upside risks.

How history informs the forecast

These are from my general knowledge of past episodes, so the figures are approximate:

- **Energy-driven inflation tends to fade if oil stabilizes.** After spikes in 1990, 2008, and 2022, headline CPI converged back toward core within roughly 6 to 18 months. In 2008 it went negative within a year of the peak.
- **It becomes dangerous when it spreads.** In 1973-74 and 1979-80, oil shocks combined with rising inflation expectations and slow central-bank action produced double-digit inflation. Today's 3.6% expectations and sticky core PCE are why the Fed is hiking rather than looking through the shock.
- **Base effects matter.** Oil spiked in March-May 2026 (Brent peaked near $118), so those months will be a high comparison point. Even if oil stays near $90-100, year-over-year energy inflation should fade sharply by spring 2027. The EIA expects Brent near $105 in Q4 2026 and around $74 on average in 2027 as supply recovers.
- **Core inflation is slow-moving.** Historically it rarely falls more than about 1 percentage point in a year without a recession or an unusually decisive policy response.

Base case (next 6-12 months)

- **Headline CPI:** probably peaks around 3.5-4% in Q4 2026 or Q1 2027, then declines, ending the window roughly in the 2.5-3.3% range if oil eases as the EIA expects.
- **Core CPI:** roughly 2.4-2.9%, drifting only modestly.
- **Core PCE:** easing from about 3% toward 2.5-2.8%, consistent with the Fed's own path.

These ranges are my rough synthesis of the data above, not published forecasts.

What could push inflation higher

- **Renewed oil disruption.** The EIA says Hormuz-related shut-in production averaged 4.8 million barrels a day in September, so a setback could re-accelerate energy and shipping costs, and eventually food.
- **Tariff pass-through.** The administration rebuilt its global tariffs under Section 301 after the Supreme Court struck down the IEEPA tariffs, and producer prices excluding food and energy suggest cost pressure is still building.
- **Expectations de-anchoring.** If households and firms come to expect 3-4% inflation, wage and price setting can make it self-reinforcing.
- **Fiscal and yield pressures.** Investors are wary of rising debt and deficits, and any stimulus or "checks" would add demand.
- **Tight labor.** The Fed projects unemployment staying around 4.1%.

What could push inflation lower

- **Oil normalization** from a durable Iran deal or faster supply recovery.
- **Base effects** from the spring 2026 energy spike.
- **Fed tightening and weaker demand.** The weak jobs report and consumer confidence at a 12-year low point to softening spending.
- **Shelter disinflation.** Shelter inflation has eased to about 3.0%.
- **Tariff refunds or court rulings** that cut effective tariff rates, which are being challenged in court.

Scenarios (illustrative, not probabilities)

Scenario	Conditions	Headline CPI by late 2027
Disinflation	Oil falls toward $70-80, Fed holds, demand softens	~2.2-2.8%
Muddle-through (base)	Gradual oil easing, sticky core	~2.5-3.3%
Re-acceleration	Renewed supply shock, tariff pass-through, expectations drift up	3.8-5%+

Uncertainty

- **Forecasts get revised a lot.** In March the Fed's median core PCE projection for 2026 was 2.7%. By September it was 3.4%, a 0.7-point upward revision in six months. Typical 12-month-ahead forecast errors are around a percentage point even in calm periods (a rough figure from memory), and larger during shocks.
- **Measurement gaps.** Core CPI at 2.4% and core PCE at roughly 3% differ unusually, and a methodology change in PCE is expected to mechanically lower the reading, which complicates comparisons.
- **Policy is unpredictable.** Tariff authorities, Fed leadership, and fiscal plans have changed repeatedly, and the Fed itself is divided on 2027, with some officials seeing cuts and others seeing more hikes.
- **Geopolitical dependence.** Oil is the biggest swing factor, and it can move $15-30 on a single news event.
- **Data limits.** I relied on news and agency summaries, some of which conflict on exact figures. Check the latest CPI, PCE, and Fed releases before relying on any number here.

I'm not a financial advisor, and decisions about investing, borrowing, or contracts should consider your own circumstances.

Figure A7. ChatGPT response - Section 3: Risk Identification and Mitigation

A7 ChatGPT Risk Analysis

Over the next 6 to 12 months, the most important financial risks appear to be market volatility, credit risk, and liquidity risk, with market volatility currently presenting the greatest immediate concern.

1. Market volatility — High severity
Volatility could remain elevated because of interest-rate uncertainty, energy-price shocks, geopolitical conflict, trade-policy changes, and concentrated market leadership. Sharp movements in stocks, bonds, or commodities could reduce portfolio values and increase losses for investors and financial institutions.

Mitigation: Diversify across asset classes, avoid excessive concentration in one sector, use stress testing, and maintain appropriate hedging strategies where suitable.

2. Credit risk — Moderate to high severity
Higher borrowing costs and slower economic growth could make it harder for businesses and consumers to repay debt. Companies with weak cash flow or heavy debt loads may be especially vulnerable if interest rates stay elevated.

Mitigation: Strengthen credit screening, monitor borrower quality closely, reduce exposure to highly leveraged borrowers, and maintain adequate loan-loss reserves.

3. Liquidity risk — Moderate severity
Periods of market stress can make some assets harder to sell without accepting large losses. Financial institutions may also face pressure if customers withdraw funds quickly or if short-term funding becomes more expensive.

Mitigation: Maintain strong cash reserves, hold highly liquid assets, diversify funding sources, and conduct regular liquidity stress tests.

Overall, market volatility ranks as the highest near-term risk, followed by credit risk and then liquidity risk. However, these risks are connected. For example, a sharp market decline could weaken borrower finances and create liquidity pressure at the same time. The severity of each risk could change quickly depending on interest rates, energy prices, economic growth, and geopolitical events, so these rankings should be treated as scenario-based rather than guaranteed.

Figure A8. Claude response - Section 3: Risk Identification and Mitigation

Based on the earlier analyses, the biggest risks come from the interaction of a hawkish Fed, 5%+ long-term Treasury yields, an oil shock, and a record-high, narrowly led stock market. I rank them below by potential damage and breadth, not by probability. None of these is a prediction, and most have no reliable probability estimate.

Ranking at a glance

Rank	Risk	Category	Potential severity
1	Long-end bond selloff straining Treasury market liquidity	Liquidity	Very high
2	Private credit defaults and redemption gating spilling into banks and insurers	Credit / liquidity	High
3	Rate- and valuation-driven equity drawdown	Market volatility	High
4	Renewed oil or geopolitical shock	Market volatility	High, but binary
5	Consumer and leveraged-corporate credit deterioration	Credit	Moderate to high, slower
6	Bank funding and balance-sheet strain	Liquidity / credit	Moderate (evidence dated)

1. Treasury market stress (liquidity)

Evidence. September was one of the sharpest bond repricings in years: the 10-year yield rose 54 bps to 5.27%, the 30-year reached its highest level in roughly two decades, and rate volatility stayed above April's highs. The Treasury index is down about 2.7% for the year. Hedge fund leverage in Treasury relative-value trades has grown, and in March 2020 and April 2025 Treasury liquidity deteriorated faster than expected.

Who it hits. Leveraged funds (margin calls, forced selling), dealers, banks and insurers holding long bonds, and any investor using Treasuries as collateral. Because Treasuries price everything else, stress here transmits to stocks and credit.

Mitigation.

- Institutions: stress-test collateral and margin needs against another 50-100 bp move, hold unencumbered high-quality liquid assets, diversify funding, and set limits on repo and basis-trade leverage.
- Investors: keep duration moderate, keep cash or T-bills for needs over the next 12 months, avoid leverage on bond positions, and ladder maturities.

2. Private credit stress (credit and liquidity)

Evidence. Fitch put the private credit default rate at a record 6.3%, but measures range widely (about 1% to 19%) depending on definitions. In 1Q26, redemption requests at non-traded BDCs averaged 9-10% of net asset value, well above the 5% limits many set. Banks are exposed indirectly through fund financing, Jamie Dimon warned losses will be higher than expected, and the Fed has queried major banks. Software-sector exposure is a particular worry, and one bank modeled a severe case with 14-15% defaults. Counterpoints: public high-yield spreads are still near historic tights, and some analysts see no systemic stress so far.

Who it hits. Retail and semi-liquid fund investors (gated redemptions, markdowns), BDC shareholders, insurers, and banks with leveraged-lending relationships.

Mitigation.

- Institutions: map look-through exposure to private credit funds, stress-test valuation marks, tighten covenants, and diversify away from concentrated software and PIK-heavy borrowers.
- Investors: size private credit as money you won't need for years, read redemption terms, check manager valuation practices, and avoid concentration in one sector or manager.

3. Equity drawdown from rates and valuation (market volatility)

Evidence. The S&P 500 hit a record 7,818.93 on Oct 6 while breadth has narrowed, with only about a quarter of stocks above a key trend level. Goldman notes AI beneficiaries account for roughly half of earnings growth, and the Fed's median path implies no cuts in 2027. History is mixed: in 2022 stocks and bonds fell together, and in 2023 a rise in yields toward 5% coincided with a roughly 10% pullback before a rebound (from my memory, so approximate).

Who it hits. Equity-heavy investors, index funds, and anyone whose portfolio is concentrated in a few large tech names.

Mitigation.

- Institutions and investors: rebalance toward target weights, diversify beyond the largest AI-linked names, use options collars or put spreads sparingly for downside protection, and avoid leverage.
- Investors: match stock exposure to your time horizon, since pullbacks of 10% occur roughly once a year on average.

4. Oil and geopolitical shock (market volatility)

Evidence. Brent is about $101, 54% above a year ago, and the EIA projects about $105 in Q4. Oil has swung $15-30 on single events, and renewed disruption would hit inflation, yields, and equities together.

Who it hits. Energy-sensitive sectors (airlines, chemicals, transport), consumers, and any portfolio that assumes inflation will fall.

Mitigation.

- Institutions: hedge fuel and input costs with swaps or futures, and run scenario analysis on $120+ and $70 oil.
- Investors: hold some real-asset or energy exposure, keep inflation-protected bonds in the mix, and avoid assuming a rapid oil decline.

5. Consumer and leveraged-corporate credit (credit)

Evidence. Credit card delinquencies were elevated and auto defaults rising in a late-2025 outlook, consumer confidence recently fell to a 12-year low, and energy costs are squeezing budgets. High-yield returned about -2.5% in September as spreads widened to roughly 306 bps, still tight historically, which could mean credit is underpricing risk.

Who it hits. Consumer lenders, auto and card ABS holders, and leveraged companies refinancing at higher rates.

Mitigation.

- Institutions: tighten underwriting, build loan-loss reserves, and monitor early-stage delinquencies.
- Investors: favor higher-quality credit, avoid reaching for yield, and diversify issuers.

6. Bank funding and balance-sheet strain (liquidity and credit)

Evidence. Banks carried elevated unrealized securities losses and weakened commercial real estate asset quality through 2024 (the latest regulator data I found is dated). With 10-year yields now higher than in 2024, those pressures likely haven't eased, though that is my inference, not a measured figure.

Who it hits. Regional banks with concentrated deposits or CRE exposure, and their uninsured depositors.

Mitigation.

- Institutions: diversify deposits, maintain borrowing capacity at the Fed discount window and FHLB, and hedge duration.
- Depositors: keep balances within insurance limits or spread across institutions.

Uncertainty

- **Rankings are judgments.** A reasonable analyst could order these differently, especially risks 2 and 3, and the risks interact. A bond selloff, for example, could trigger both equity declines and private credit redemptions.
- **Data conflicts.** Private credit default estimates vary severalfold, and some sources I used are dated or secondary.
- **Missing indicators.** I don't have current bank-level, hedge fund leverage, or repo stress data, which are exactly what would sharpen this.
- **Tail risks are rare but large.** Most of these have historically been absorbed without crisis, and some may never materialize. The point of mitigation is to survive the cases that do.

This is general analysis, not personalized advice. I'm not a financial advisor, and institutions should validate any of this against their own exposures and regulatory requirements.

Figure A9. ChatGPT response - Section 4: Optimization and Enhancement

A9 ChatGPT Optimization

The biggest weaknesses in the earlier forecasting and risk-analysis tasks were uncertainty, reliance on changing market data, and the possibility of presenting confident answers based on incomplete or outdated information. Improving LLM performance in financial applications therefore requires both better inputs and stronger review processes.

Better prompt design can improve consistency by clearly defining the financial task, time horizon, data source, required output format, and treatment of uncertainty. For example, asking a model to separate verified facts from assumptions makes it easier to evaluate the quality of the response.

Verified financial data is essential because forecasts are only as reliable as the information used. Connecting an LLM to trusted sources such as official economic reports, market data feeds, or company filings can reduce errors caused by outdated or incorrect information.

Retrieval-augmented generation (RAG) can improve reliability by allowing the model to retrieve relevant financial information before generating an answer. This helps ground responses in current documents and data instead of relying only on information learned during model training.

Human review is especially important for high-impact financial decisions. Analysts can verify calculations, challenge assumptions, and identify risks that the model may overlook. LLMs are better suited to supporting analysis than making final decisions independently.

Uncertainty reporting should also be built into the output. Models should identify assumptions, possible alternative scenarios, confidence levels, and major factors that could change the forecast. This reduces the risk of presenting a financial prediction as certain.

Even with these improvements, important limitations would remain. Financial markets are affected by unexpected political events, economic shocks, investor behavior, and rapidly changing conditions that cannot always be predicted from historical data. LLMs can also misunderstand sources, generate incorrect interpretations, or reflect bias in the information they retrieve. For these reasons, improved LLM systems can make financial analysis more useful and reliable, but they should still be combined with verified data, independent analysis, and human oversight.

Figure A10. Claude response - Section 4: Optimization and Enhancement

The earlier analyses had real weaknesses, and several fixes follow directly from them. I'll start with what went wrong, then the improvements, then what would still be left over.

Weaknesses in my earlier work

- **I didn't run the historical datasets.** You asked me to use historical S&P 500, oil, and inflation data. Instead I relied on base rates from memory (e.g., "roughly 70-75% of years are up," "annual standard deviation ~17-19%"), which I flagged as approximate but never verified or computed.
- **Conflicting and mislabeled sources.** Search results disagreed on whether the 30-year yield hit its highest level since 2002 or 2007, some CPI articles were from 2025 but labeled 2026, and private credit default estimates ranged from 1% to 19%. I could flag these but not resolve them.
- **Secondary, dated data.** I used news summaries and a regulator report on bank losses that was a year old.
- **Judgment dressed as analysis.** The scenario ranges and risk rankings were my synthesis, not outputs of a tested model.
- **No track record.** I couldn't check whether similar forecasts had been accurate.

Improvements

1. Verified financial data. The biggest single fix is connecting the model to authoritative sources, such as FRED, BLS, the EIA, and the Fed's published projections, through structured tools instead of web summaries. The model should pull a time series, record the source and retrieval date, and compute statistics in code rather than recalling them. This would have replaced my "from memory" base rates with actual calculations. A data-validation layer (checking units, dates, and revisions) would catch problems like the mislabeled CPI articles.

2. Retrieval-augmented generation (RAG). RAG anchors answers in specific documents, such as FOMC statements, 10-Ks, and central bank reports. Done well, it lets the system cite exact passages, prefer primary sources over aggregators, and flag when retrieved sources conflict instead of silently picking one. Useful design choices: rank sources by authority and recency, store timestamps, and require the model to say "not found in sources" instead of filling gaps.

3. Better prompt design. Prompts can enforce structure and honesty:

- Require a source and a date for every figure.
- Separate "retrieved facts," "calculations," and "judgments" in the output.
- Ask for base rates and a reference class before any prediction.
- Ask for the strongest counter-argument to each conclusion.
- Specify the horizon, the benchmark, and what would change the forecast.

A prompt like "forecast core PCE in 12 months; show last 10 years of forecast errors; give a range and the assumptions behind it" produces much better output than "will inflation fall?"

4. Tool use and quantitative modeling. For forecasting, the language model should orchestrate, not predict. It can call econometric or machine-learning models (e.g., a regression on inflation expectations, a volatility model, a scenario simulator), run backtests, and explain the results. This would turn my loosely stated ranges into calibrated ones, with documented error rates.

5. Uncertainty reporting. Good practice includes:

- Prediction intervals instead of point estimates, tied to historical error.
- Explicit scenario weights only when they come from a model or consensus, otherwise stating that no probabilities are available.
- Confidence labels on each claim (verified data, modeled estimate, informed judgment).
- Tracking forecast accuracy over time, so claimed uncertainty can be compared with actual calibration.

6. Human review. Qualified analysts should review outputs before they reach decisions, especially for regulated or high-stakes work. Review is most useful when it targets specific checkpoints: data provenance, assumption sanity, model validation, and compliance language. Reviewer feedback can also be stored and used to improve prompts and evaluation sets.

7. Evaluation and monitoring. Build test sets of past questions with known answers (e.g., "what was the 10-year yield on this date?"), measure hallucination and numerical error rates, and re-run them whenever the model or data pipeline changes. Log every output so errors can be traced.

Mapping weaknesses to fixes

Weakness	Main fix
Base rates from memory	Verified data plus code execution
Conflicting or dated sources	RAG with source ranking and conflict flagging
Unsupported scenario ranges	Quantitative models with backtests
Overconfident framing	Uncertainty reporting and calibration tracking
Subjective rankings	Structured criteria plus human review

Limitations that would remain

- **Markets are partly unpredictable.** Better tooling can't remove genuine uncertainty. Wars, policy shifts, and sentiment changes aren't forecastable from data alone, and forecasters of all kinds missed this year's oil and inflation moves badly.
- **Structural breaks.** Models learn from history, and regimes change. A hiking Fed plus an oil shock plus rebuilt tariffs has few precedents.
- **Garbage in, garbage out.** Verified data can still be revised, delayed, or mismeasured (as with the CPI/PCE divergence), and private markets like private credit have unreliable and inconsistent data.
- **Reflexivity.** If many participants use similar models, their behavior can change the outcomes being forecast.
- **LLM-specific errors remain.** Even with retrieval, models can misread tables, misattribute figures, or sound more confident than the evidence supports. Fluent writing can hide weak reasoning, and the model can't reliably know when it's wrong.
- **Human review has limits too.** Reviewers can be fatigued, over-trust fluent output, or share the model's blind spots, and review costs time and money.
- **Judgment calls don't disappear.** Risk rankings, scenario design, and what counts as "most important" involve values and priorities that no data fix can settle.
- **Accountability and compliance.** Advice-giving carries legal, fiduciary, and disclosure obligations that an AI system can't assume, so humans must remain responsible.

Practical conclusion

LLMs are best treated as research assistants that retrieve, organize, calculate through tools, and explain, with humans accountable for decisions. The combination of verified data, RAG, quantitative models, explicit uncertainty, and human review would make my earlier analyses more reliable and checkable, but not able to predict markets. Even a well-built system should keep reporting wide ranges and caveats, because that is what the evidence supports.

Appendix B: Quantitative Evaluation Results

The following tables reproduce the key generated result files from the GitHub repository.

B1. Model Summary (results/model_summary.csv)

Model	Prompt Type	Forecasts	MAPE
ChatGPT	Baseline	3	9.31%
ChatGPT	Optimized	3	3.75%
Claude	Baseline	3	10.66%
Claude	Optimized	3	7.67%

B2. Optimization Comparison (results/optimization_comparison.csv)

Model	Baseline MAPE	Optimized MAPE	Improvement	Relative Improvement
ChatGPT	9.31%	3.75%	5.56 points	59.72%
Claude	10.66%	7.67%	2.99 points	28.05%

B3. Held-Out Actual Results

Metric	Held-Out 2024 Actual
S&P 500 final trading day	5,881.63
WTI December monthly average	$70.12 per barrel
U.S. CPI December YoY	2.9%

All forecast-level calculations are available in results/forecast_evaluation.csv, including the forecast value, actual value, absolute error, and absolute percentage error for each of the twelve forecasts.

Appendix C: Reproducibility Workflow

The technical project can be reproduced from the repository using the following workflow:

- Clone or download the repository.
- Use uv with the project metadata in pyproject.toml and uv.lock.
- Review the historical inputs and held-out actuals in the data/ directory.
- Review the exact prompts and complete model outputs in artifacts/.
- Run: uv run python src/evaluate.py
- Compare the generated CSV files in results/ with the tables in this report.

Evaluation formula

Absolute Percentage Error (APE) = |Forecast - Actual| / |Actual| × 100

Mean Absolute Percentage Error (MAPE) = the arithmetic mean of the three APE values for each model and prompt condition.

Audit trail
The report, repository data files, exact model responses, scoring script, and generated outputs together provide a traceable chain from prompt to forecast to held-out actual to calculated error. This directly supports verification of the analysis described in the report.